

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Industry Problem Solving

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 – Viva Based Questions	Assessment:	Assessment Tool 1– Poster Presentation
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	B. Ganga Jayadev Reddy	Reg. No.:	192472248
Section / Batch:	CSE AI	Date:	21 - 08 - 2026

Code of Conduct

I B. Ganga Jayadev Reddy (192472248) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Ganga Jayadev Reddy

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding ; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding ; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

A. Introduction to Statistical Inference

1. What is statistical inference?

Answer: Statistical inference is the process of using sample data to draw conclusions about a population through estimation and hypothesis testing.

2. Why is statistical inference important in data science?

Answer: It helps data scientists make reliable decisions about large populations using limited sample data.

3. What is the difference between descriptive statistics and inferential statistics?

Answer: Descriptive statistics summarizes data, whereas inferential statistics uses sample data to make conclusions about a population.

4. What is a population in statistics?

Answer: A population is the complete set of individuals, objects, or observations being studied.

5. What is a sample?

Answer: A sample is a subset of the population selected for analysis.

6. Why do data scientists usually work with samples instead of complete populations?

Answer: Samples reduce cost, time, and effort while still providing accurate estimates.

7. What is a population parameter?

Answer: A population parameter is a numerical value describing a population, such as the population mean (μ) or proportion (p).

8. What is a sample statistic?

Answer: A sample statistic is a numerical value calculated from a sample, such as the sample mean ($\bar{x}$).

9. Give a real-world example where statistical inference is used in industry.

Answer: An e-commerce company estimates customer conversion rates using a sample of website visitors.

10. What are the two major activities involved in statistical inference?

Answer: The two activities are estimation and hypothesis testing.

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

Answer: It interprets probability as the long-run frequency of repeated events.

12. What is the basic idea behind frequentist statistics?

Answer: Population parameters are fixed, and conclusions are based on repeated random sampling.

13. How does the frequentist approach differ from the Bayesian approach?

Answer: Frequentist treats parameters as fixed, while Bayesian treats parameters as random variables with prior information.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

Answer: The population parameter is fixed, while the sample data is random.

15. What does long-run frequency mean in frequentist statistics?

Answer: It is the proportion of times an event occurs after many repeated experiments.

16. Why is random sampling important in the frequentist approach?

Answer: It reduces sampling bias and improves representativeness.

17. Can a population parameter be considered a random variable? Why?

Answer: No. In the frequentist approach, the parameter is fixed but unknown.

18. Give an example of a frequentist problem in data science.

Answer: Testing whether a new machine learning model significantly improves prediction accuracy.

C. Point Estimation

19. What is a point estimate?

Answer: A point estimate is a single value used to estimate a population parameter.

20. What is an estimator?

Answer: An estimator is a statistical formula used to calculate an estimate from sample data.

21. What is the difference between an estimator and an estimate?

Answer: An estimator is the method; an estimate is the calculated value.

22. What is the point estimator for the population mean?

Answer: The sample mean ($\bar{x}$).

23. How do you estimate a population proportion from a sample?

Answer: Use the sample proportion: $\hat{p} = x/n$.

24. What is the point estimate of population variance?

Answer: The sample variance (s^2).

25. Why is the sample mean commonly used?

Answer: It is unbiased, efficient, and easy to calculate.

26. What properties should a good estimator have?

Answer: It should be unbiased, consistent, efficient, and sufficient.

27. What is an unbiased estimator?

Answer: An estimator whose expected value equals the true population parameter.

28. What is meant by the bias of an estimator?

Answer: Bias is the difference between the expected estimate and the true parameter.

29. What is the difference between bias and variance?

Answer: Bias measures systematic error, while variance measures estimate variability.

30. Why can two different samples produce different point estimates?

Answer: Different random samples naturally contain different observations.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Answer: Because it is calculated from a sample instead of the entire population.

32. What is the sampling distribution of an estimator?

Answer: It is the probability distribution of an estimator over repeated samples.

33. What is the standard error?

Answer: The standard deviation of a sampling distribution.

34. How is standard error different from standard deviation?

Answer: Standard deviation measures data spread, while standard error measures estimate precision.

35. How does increasing the sample size affect the standard error?

Answer: Increasing sample size decreases the standard error.

36. Why does variability decrease when sample size increases?

Answer: Larger samples produce more stable and reliable estimates.

37. What is the relationship between standard error and confidence intervals?

Answer: A smaller standard error produces a narrower confidence interval.

38. If two samples have different sample means, does it mean one is incorrect?

Answer: No. Different samples naturally produce different means due to sampling variation.

E. Confidence Intervals

39. What is a confidence interval?

Answer: A confidence interval is a range of values likely to contain the true population parameter.

40. Why is a confidence interval more informative than a point estimate?

Answer: It provides both the estimate and its uncertainty.

41. What does a 95% confidence level mean?

Answer: In repeated sampling, about 95% of confidence intervals would contain the true parameter.

42. What is the difference between a confidence level and a confidence interval?

Answer: The confidence level is the reliability percentage, while the confidence interval is the calculated range.

43. What factors determine the width of a confidence interval?

Answer: Sample size, variability, and confidence level.

44. How does increasing the sample size affect the width of a confidence interval?

Answer: It makes the interval narrower.

45. How does increasing the confidence level from 95% to 99% affect the interval?

Answer: The interval becomes wider.

46. What is the difference between a z-confidence interval and a t-confidence interval?

Answer: Use z for known SD or large samples; use t for unknown SD and small samples.

47. When would you use the t-distribution?

Answer: When the population standard deviation is unknown and the sample size is small.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Answer: A statistical method used to evaluate whether sample evidence supports a population claim.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

Answer: H_0 states no effect or difference, while H_1 states there is an effect or difference.

50. How can a confidence interval be used to make a decision about a hypothesis?

Answer: If the claimed value lies inside the confidence interval, do not reject H_0 ; otherwise, reject H_0 .